

Learning from One-Shot Human Video for Cross-Embodiment Optimization and Reasoning

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Abstract—Robot learning is frequently bottlenecked by the scarcity of high-quality, robot-specific demonstration data. While existing learning-from-video methods attempt to leverage human demonstrations, they typically rely on complex, multi-stage engineering pipelines that are highly susceptible to compounding errors and struggle with cross-embodiment transfer. Vision-Language Model (VLM)-driven Code as Policy (CaP) frameworks offer a robust alternative to these issues. However, their reliance on expert level, manually engineered, task-specific prompts limits the scalability. In this paper, we propose a novel pipeline that automatically extracts key geometric constraints, state changes, and task-relevant information from a single, non-expert human video demonstration in the wild. This extracted context is used to dynamically synthesize near-optimal prompts for the CaP framework, effectively bridging the embodiment gap without requiring iterative prompt engineering. We evaluate our method across three distinct robotic manipulation tasks. Experimental results demonstrate that our approach significantly outperforms purely agentic baselines in both task completion rates and reward metrics. Furthermore, ablation studies validate the necessity of our structured state-extraction and templated prompting components for achieving reliable task execution.

Index Terms—Learning from Videos, Robotics Learning, Code as Policy

I. INTRODUCTION

GENERAL-purpose manipulation remains a core challenge in Robotics. The generalization of such systems is severely bottlenecked by the scarcity of high-quality, robot-specific demonstration data [1]. Traditional data collection paradigms, such as physics-based simulation environments, teleoperation, and wearable devices, have shown efficacy in specific scenarios but face limitations. Simulations suffer from a significant sim-to-real gap [2]. Teleoperation gives the best quality robot data, yet it is notoriously difficult to scale, costly, embodiment-dependent [3, 4], and unintuitive [5]. Wearables often introduce a visual embodiment gap during data collection [6], or need special design to match the robot [5, 7]. Consequently, extracting policies directly from in-the-wild human demonstration videos has emerged as a highly scalable and promising alternative.

Despite the promise of learning from videos, existing methods heavily rely on precise and accurate frame-by-frame engineering pipelines. These approaches typically focus on dense depth estimation, 3D reconstruction [8], and continuous motion retargeting [9]. Such tightly coupled architectures suffer from severe mutual dependency. Minor errors in upstream visual perception cascade and amplify during downstream action generation, ultimately resulting in severe jittering trajectories during execution. Furthermore, existing

methods often require "human videos" to be unnaturally performed by operators accommodating the robot's kinematics, rather than utilizing true unstructured, in-the-wild data [8–10]. This raises a fundamental question: To enable a robot to successfully complete a manipulation task, do we genuinely require continuous demonstration trajectories for every single second?

Recently, Vision-Language Model (VLM)-driven Code as Policy (CaP) frameworks have offered a novel perspective to address these issues [11, 12]. By generating executable code for spatial reasoning and action planning, CaP bypasses traditional dense trajectory planning. However, CaP encounters a significant bottleneck in practical applications: it heavily relies on meticulously fine-edited oracle prompts to describe the initial state and geometric constraints of the task. Without this exhaustive information, purely agentic approaches frequently fail, severely limiting the real-world applicability of CaP.

To overcome this challenge, we propose a novel one-shot cross-embodiment transfer pipeline. Instead of pursuing dense trajectory matching, our approach extracts key operational states and geometric constraints from a single, non-expert, in-the-wild human video. By integrating this sparse yet critical information with predefined templates, we automatically synthesize near-optimal prompts to directly guide the CaP system in task execution.

The core contributions of this paper are summarized as follows:

- We propose a lightweight cross-embodiment transfer pipeline that automatically extracts key states and geometric constraints from a single human demonstration video. This replaces the fragile frame-by-frame reconstruction and retargeting mechanisms in traditional Learning from Videos, significantly reducing the mutual dependency between system modules.
- We design an automated CaP prompt synthesis mechanism that combines extracted key states with predefined templates to generate high-quality task descriptions, eliminating the reliance on tedious manual prompt engineering.
- We provide comprehensive empirical evaluation and failure analysis. In manipulation tasks such as Bottle Knock and Wipe, our method significantly outperforms the purely agentic baseline in both completion rates and reward metrics.

II. RELATED WORK

A. Learning from Human Videos

Directly extracting manipulation policies from human demonstration videos presents a scalable alternative to traditional data collection methods. However, a significant portion of existing literature relies heavily on tightly controlled laboratory settings. Methods such as those proposed in [13] and [9] attempt to explicitly map continuous human hand motions to specific robotic kinematics. This frame-by-frame, dense retargeting approach is highly susceptible to perception errors. They often cause jittering movements during simulation or physical robot execution.

More recently, as discussed in the seminar, works like [8, 10] have advanced the learning-from-video pipeline by incorporating sophisticated spatial tracking and 3D reconstruction mechanisms. Despite these architectural improvements, this pipeline still necessitates highly precise depth estimation. Such strict perceptual prerequisites severely constrain the method’s applicability to true in-the-wild videos, as noisy real-world environments and dynamic lighting inevitably degrade depth estimation quality.

Furthermore, these paradigms fundamentally assume that the observed human action is always optimal. In natural, unstructured environments, human demonstrations are often suboptimal, heuristic, or contain redundant motions. Forcing a robot to densely replicate such trajectories is inefficient. In contrast, our approach circumvents the need for precise continuous tracking and the assumption of demonstrator optimality by extracting only sparse, task-critical states to guide policy generation.

B. Cross-Embodiment Transfer

Transferring manipulation skills from humans to robots introduces a profound embodiment gap, as distinct hardware morphologies dictate fundamentally different physical interactions. Even identical high-level actions can yield completely different environmental effects depending on the robot’s physical constraints. To bridge this gap, some approaches initially focus on perceptual alignment. For instance, DexUMI [6] addresses the visual mismatch by employing post-processing and image inpainting techniques, effectively superimposing the target robotic end-effector in the video. However, it masks only the visual discrepancy but does not resolve the mechanical and kinematic mismatch between the human hand and the robot hardware.

To explicitly address the physical mapping of actions across morphologies, frameworks such as HoMMI [14] attempt to translate unconstrained human demonstrations into robot-executable whole-body mobile manipulation trajectories. Crucially, HoMMI utilizes a “Look-At” focal point as a spatial anchor, aligning the demonstrator’s visual attention with the robot’s physical workspace. Yet the underlying methodology fundamentally remains a motion-mimicking paradigm.

Recognizing the limitations of strict kinematic alignment, recent paradigms argue for decoupling the policy from rigid, embodiment-specific continuous control spaces. As demonstrated by LAP [15], language-action pretraining provides a

mechanism to abstract the action space entirely. Such a method enables the system to reason about tasks at a semantic and intentional level, achieving successful policies on different robots.

C. Vision-Language Models and Code as Policy

The integration of AI, Large-Language Models [16], and Vision-Language Models into embodied AI has catalyzed a paradigm shift in robot learning, enabling systems to leverage vast, internet-scale pretraining for robotic tasks. Landmark architectures such as RT-2 [17] have demonstrated remarkable zero-shot semantic generalization. However, while these VLM-based approaches excel at high-level semantic reasoning and visual recognition, they often struggle with the precise spatial awareness and geometric constraints required for fine-grained manipulation.

To bridge the gap between abstract semantic reasoning and precise physical execution without relying on dense trajectory mapping, the Code as Policy [11, 12] framework has emerged as a robust alternative. Instead of predicting continuous motor commands, the CaP pipeline utilizes large language models to autonomously synthesize executable control programs (Scripts invoking robot APIs) based on environmental context and task instructions.

Despite its powerful spatial reasoning capabilities, the empirical success of the CaP framework is fundamentally bottlenecked by its input. To achieve optimal performance, CaP systems require meticulously crafted prompts that explicitly describe precise initial states, end-effector orientations, and strict geometric boundaries. Our proposed pipeline automatically extracts these critical physical states from a single human video to synthesize near-optimal prompts, reducing the need for human-in-the-loop prompt engineering.

III. METHODS

Our proposed methodology introduces a one-shot, cross-embodiment transfer pipeline designed to generate zero-shot executable robotic policies from a single, in-the-wild human demonstration. As illustrated in Fig. 1, the framework deliberately bypasses dense motion retargeting. Instead, it processes the RGB video through a decoupled, three-stage architecture: object-centric visual processing, acceleration-based keyframe extraction, and automated prompt generation. The extracted sparse keyframes, representing critical state transitions, are systematically combined with the robot’s initial spatial observation to synthesize a highly structured task description. This description serves as a near-optimal prompt for a VLM operating within a CaP-X [11] architecture, generating executable Python control scripts.

A. Visual Processing

Inspired by related work [15], to abstract the human demonstrator’s intent without conforming to human kinematic priors, our system tracks the manipulated objects rather than the human hand. Given a video, we employ the Segment Anything Model (SAM) to generate precise object masks across all



Fig. 1. Pipeline of Our Method. Top left figure: initial human video demonstration. Top center: results after video post-processing and centroid trajectory visualization. Right column: extracted key frames. Bottom center 2 figures: initial observation in the global camera and the wrist camera. Bottom left: Cap-X backbone [11].

frames. From these masks, we continuously calculate the 2D centroid of the target object to represent its planar trajectory. To elevate this planar data into a spatial trajectory, we integrate FoundationStereo [18] for depth estimation, projecting the 2D centroids into a continuous 3D trajectory sequence. This process yields a lightweight, object-centric representation of the task execution, independent of the demonstrator’s embodiment.

B. Keyframe Detection

Continuous 3D trajectories derived from human videos frequently contain heuristic adjustments, redundancies, and non-optimal paths that are detrimental to rigid robotic execution. To isolate the fundamental task components, we dynamically identify sparse keyframes corresponding to critical state changes, inspired by [10]. We compute the 3D acceleration of the object’s centroid trajectory across the temporal axis. Significant dynamic **object** shifts manifest as acceleration peaks. Frames corresponding to these local extrema are extracted and chronologically sequenced. This acceleration-based temporal down-sampling ensures that only the most task-critical geometric transformations are preserved, significantly mitigating the cascading errors typical of dense mutual-dependency pipelines.

C. Generating Task Description from Keyframes and Initial States

The final stage bridges the gap between the extracted visual states and the robot’s constrained action space. We formulate

a structured prompt template that translates the sequenced keyframes and the robot’s current environmental observations into a comprehensive task description for the CaP-X agent. The synthesis is guided by four reasoning modules:

- **Scene Description:** Grounds the dual-view observations into a strict physical layout. It establishes an absolute reference coordinate system and abstracts objects into pure geometries and bounding boxes, deliberately ignoring task-irrelevant visual features unless mathematically necessary for object disambiguation.
- **State Change & Goal Reasoning:** Isolates the fundamental task objective by analyzing the physical delta between the initial state and the final keyframe. It explicitly disregards the sub-optimal trajectory taken by the human demonstrator. Instead, it translates visual goals into algorithmic primitives (e.g., asking the robot to “sweep the min/max X and Y bounds of object A” rather than providing vague semantics like “clean the board”).
- **Constraints Formulation:** This module establishes rigorous spatial and physical boundaries to ensure execution feasibility, divided into two distinct categories. *Task Constraints* explicitly encode the relational semantics between distinct objects or within the intrinsic geometry of a single object. Rather than merely defining end-effector poses, this sub-module enforces the necessary topological conditions for task success. This includes interpreting mandatory contact states between entities, establishing strict relative elevations (e.g., requiring one object to be positioned strictly above another), and preserving part-

level spatial dependencies. *Environment Constraints* govern the operational limits defined by the static context of the workspace. This involves identifying immutable environmental elements and bounding the physical support manifold, actively preventing terminal execution failures caused by unsupported interactions, such as inadvertently pushing a target object beyond the spatial boundaries of the table.

- **Suggested Methods:** Generates a prescriptive, step-by-step robotic execution plan that prioritizes machine efficiency over biological mimicry. It enforces the decomposition of large motions into granular waypoints and explicitly defines the exact expected contact points and approach vectors for the gripper.

By populating this template, the prompt generator automatically synthesizes the necessary contextual grounding, enabling the Cap-X framework to reason spatially and generate robust, executable code without human-in-the-loop prompt tuning.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

All experimental evaluations are conducted within the robosuite simulation framework, powered by the MuJoCo physics engine. We deploy a Franka Emika Panda manipulator equipped with a standard parallel-jaw gripper. To rigorously test the spatial reasoning capabilities of our pipeline, the system relies exclusively on a global, third-person camera for task execution. The wrist-mounted camera is solely for initial state observation and spatial grounding. We evaluate our method across three distinct manipulation tasks, each designed to stress-test specific algorithmic capabilities (illustrated in Fig. 2).

1) *Task 1: Bottle Knock:* This task requires the manipulator to strike a vertically standing bottle, tipping it horizontally onto the workspace. This explicitly tests *cross-embodiment transfer*. In the in-the-wild demonstration, the human utilizes a highly dexterous “flicking” finger motion, an action kinematically impossible for a parallel-jaw gripper. Furthermore, this task tests the system’s ability to *optimize* over human demonstrations. The raw human video contains suboptimal vertical displacement (the bottle jumping upon impact), which the robotic policy must deliberately avoid to prevent accidental grasping. The evaluation metric is a binary success reward, $\mathcal{R} \in \{0, 1\}$, indicating whether the bottle is successfully knocked down.

2) *Task 2: Wipe Spill:* The robot is tasked with grasping an eraser and clearing a simulated spill on a tabletop. This task evaluates the pipeline’s capacity for *trajectory optimization and intent extraction*. The human demonstration exhibits highly erratic, redundant motions, spending over 20 seconds repeatedly wiping a small area. The objective is to distill the core intent (maximizing coverage) without mimicking the human’s inefficient, continuous spatiotemporal trajectory. Performance is measured by a continuous reward function, scaling linearly from 0 to 1 based on the percentage of the spill successfully wiped.

3) *Task 3: Bottle Flip:* Representing the most dynamically and kinematically complex challenge, this task requires the manipulator to flip an upside-down resting bottle into an upright position. The human demonstrator utilizes complex wrist articulation with high degrees of freedom to achieve the rotation. For a robotic manipulator, naively tracking this rotational trajectory easily leads to Inverse Kinematics solver failures and reaching joint limits. This task rigorously tests the effectiveness of our environmental and task constraints. The reward is a normalized continuous metric defined by the angular displacement of the bottle, formulated as $\theta/180^\circ$, scaled linearly within $[0, 1]$.

B. Baselines

We compare our approach against a *Purely Agentic Baseline*. In this setup, we provide the raw video demonstration directly to a state-of-the-art Vision-Language Model (Gemini Pro) and prompt it to generate the task description end-to-end, completely omitting our explicit state-extraction modules, keyframe sequencing, and templated constraint reasoning.

C. Ablation Study

To quantitatively validate the contribution of the individual modules within our structured prompt generation template, we conduct a detailed ablation study. While the *Scene Description* and *State Change & Goal Reasoning* modules are preserved as the fundamental semantic baseline, we evaluate two specific variants:

- **Ablation 1 (w/o Suggested Methods):** Removes the explicit, step-by-step robotic execution plan, testing the VLM’s zero-shot ability to decompose tasks logically without guidance.
- **Ablation 2 (w/o Constraints):** Removes both Task and Environment constraints, testing the system’s susceptibility to spatial boundary violations and IK failures when physical limitations are not explicitly defined.

D. Quantitative Results

Table I shows the quantitative results of our experiments. The code generation metric determines if the code generated by VLM is executable. Average reward measures the partial success in attempting to finish the task. Task completed is a binary value that judges whether the robot finishes the task or not. From both the table I and Figure 3, our pipelined method outperforms the baseline and almost reaches the oracle.

V. DISCUSSION

A. Methodological Efficacy and the Role of Representation

Our experimental results validate that the proposed pipeline outperforms purely agentic baselines. From Fig 3, we observe a 139% average improvement in average rewards across 3 tasks. The core takeaway is that Vision-Language Models, while possessing strong common-sense reasoning, fundamentally lack implicit spatial grounding and physical intuition. By explicitly structuring the task representation around sparse

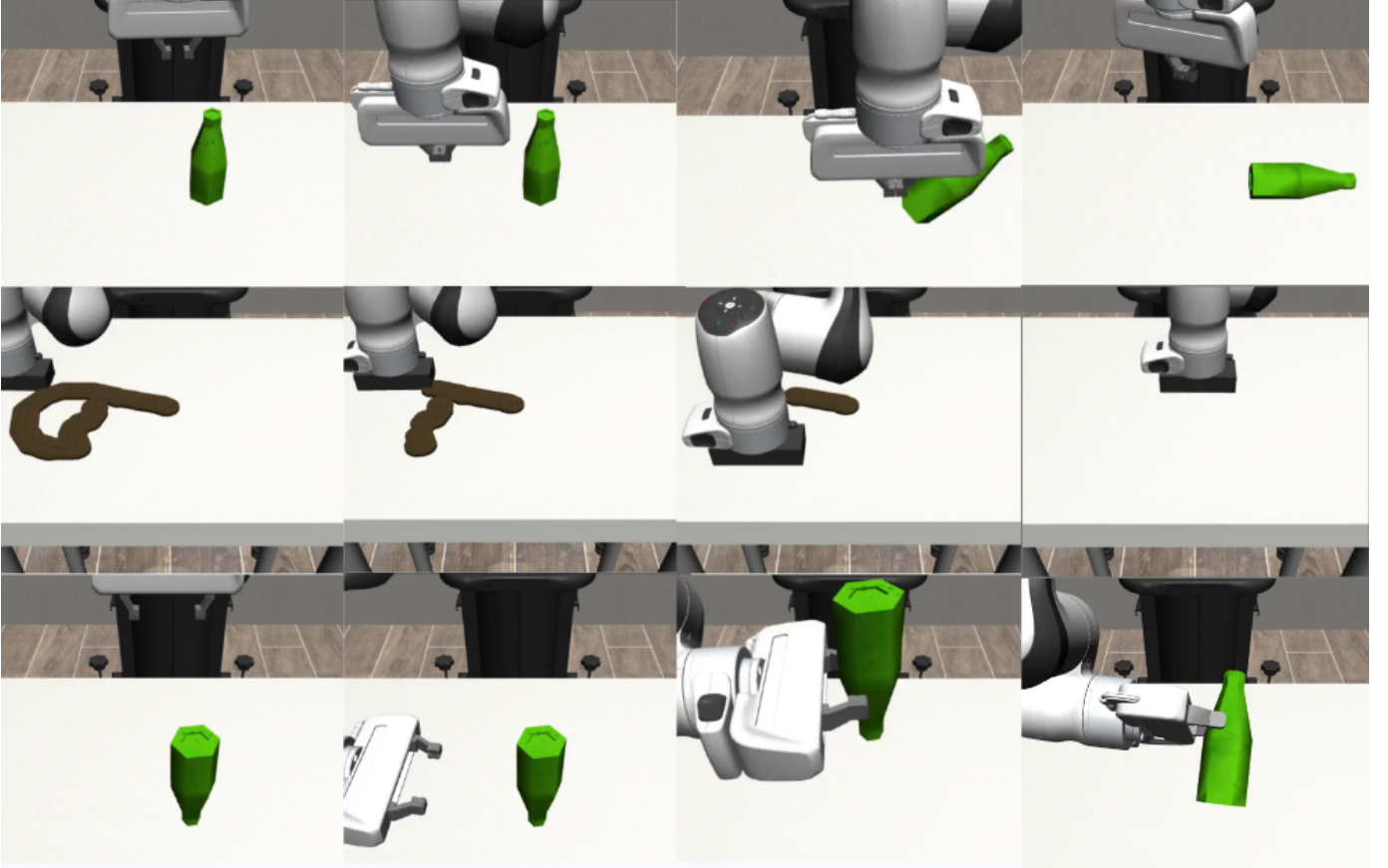


Fig. 2. Evaluation environments in the robosuite simulation. The selected tasks evaluate different axes of cross-embodiment transfer and optimization. Top Row: Bottle Knock. Middle Row: Wipe Spill. Bottom Row: Bottle Flip.

TABLE I

QUANTITATIVE RESULTS ACROSS DIFFERENT TASKS AND METRICS. WE EVALUATE CODE GENERATION SUCCESS RATE, AVERAGE REWARD, AND TASK COMPLETED. BOLD FONT INDICATES THE BEST PERFORMANCE AMONG NON-ORACLE METHODS.

Method	Code Gen. Success Rate $\uparrow$			Average Reward $\uparrow$			Task Completed $\uparrow$		
	Knock	Wipe	Flip	Knock	Wipe	Flip	Knock	Wipe	Flip
Oracle	1.000	1.000	1.000	0.900	1.000	0.668	9	10	1
Pure Agent	0.900	0.800	1.000	0.600	0.076	0.467	6	0	0
Ablation 1 (w/o Suggested Methods))	1.000	1.000	1.000	0.800	0.129	0.543	8	0	0
Ablation 2 (w/o Constraints)	1.000	1.000	1.000	0.800	0.700	0.611	8	1	0
Pipeline (Ours)	1.000	1.000	1.000	0.900	0.994	0.840	9	9	0

keyframes, geometric boundaries, and strict spatial constraints, our method effectively bridges this embodiment gap. The ablation study further corroborates this: removing predefined constraints or suggested robotic execution methods resulted in immediate performance degradation. This proves that our specific choice of representation is fundamentally necessary for reliable Code-as-Policy execution.

B. Failure Case Analysis

Despite the overall success, our pipeline encountered notable failure modes that highlight the boundaries of open-loop VLM planning.

- **Kinematic Singularities in Complex Articulation:** In the Bottle Flip task, the robot frequently failed due to

reaching joint limits. The human demonstrator executes a high-degree-of-freedom wrist roll. When the CaP system naively attempts to map this rotation to a target end-effector pose, it often generates a single, large rotational displacement that causes the Inverse Kinematics (IK) solver to fail. This indicates that for dynamically complex tasks, the VLM must be explicitly prompted to decompose rotations into finer, multi-step granular waypoints.

- **Unobservable State Dependencies:** Failures also occurred when task-critical context was visually occluded or implicitly assumed. For instance, in the Wipe Spill task, the system occasionally struggled if it could not visually infer that the eraser was already attached to the gripper. Our purely vision-centric representation falls

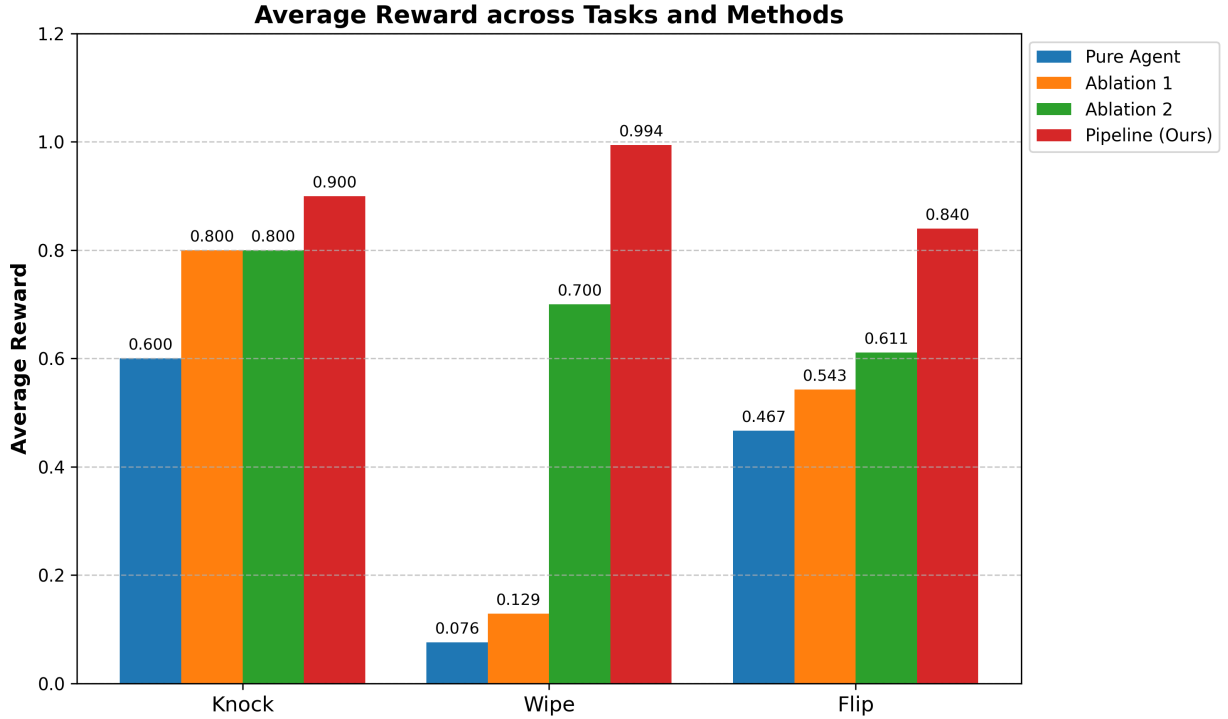


Fig. 3. Average Reward Comparison between 3 tasks

short when necessary pre-conditions are unobservable, requiring additional semantic priors to be injected into the prompt.

C. Limitations

First, all evaluations were conducted within the MuJoCo-based robosuite simulation. This inherently introduces a sim-to-real gap, as the simulation may not perfectly model the complex frictional dynamics and contact compliance present in physical execution. Validating this framework on a physical manipulator is a necessary next step. Second, our visual processing stage currently relies on stereo camera setups (RealSense) to generate accurate depth data. This limits the pipeline’s applicability to truly arbitrary internet videos. Future iterations must ablate the reliance on high-fidelity depth data, potentially integrating monocular depth estimators to enhance real-world robustness.

D. Future Work

Future research will focus on transitioning this framework from an open-loop generator to a closed-loop reasoning system. If the robotic execution fails, such as hitting a joint limit or failing an IK solve, the system should feed this runtime error back to the VLM as environmental feedback, enabling autonomous replanning. Additionally, explicitly providing the CaP agent with real-time kinematic data, including current joint configurations and strict mechanical limits, would empower the VLM to generate inherently safer and more kinematically feasible trajectories.

VI. CONCLUSION

In this paper, we introduced a novel, one-shot pipeline that translates unstructured, in-the-wild human demonstration videos into executable robotic policies. By replacing fragile, dense motion retargeting with extracting sparse keyframes and critical geometric state changes, our approach significantly mitigates compounding perception errors. We dynamically synthesize structured, near-optimal prompts that guide a VLM-driven Code-as-Policy framework, explicitly addressing the cross-embodiment gap through algorithmic constraints and spatial reasoning. Empirical evaluations in simulation demonstrate that our method outperforms purely agentic approaches in task completion and reliability. Ultimately, this work illustrates that successful robot learning from human videos does not require rigidly replicating the continuous human trajectory, but rather understanding and constraining the fundamental semantics of the task.

You can find the code of this project here: [Learning from Human Video](#)

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